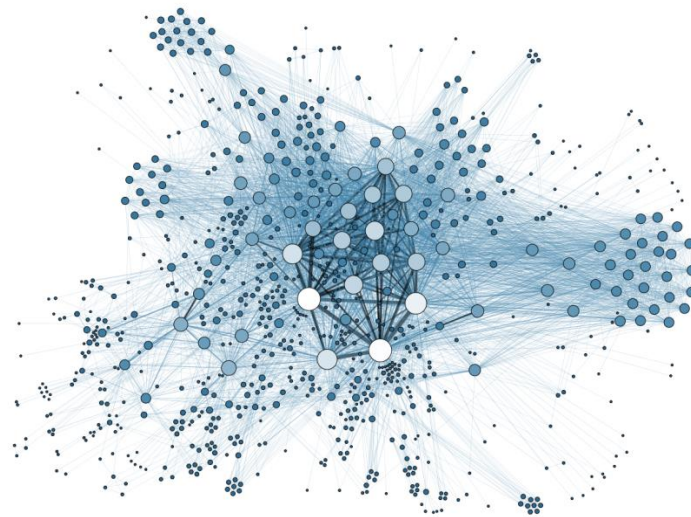


<Lab Seminar>

“Beyond Utility: The Critical Role of Fairness in AI Models!”

Htet Arkar
Undergraduate
School of Computer Science and Engineering
Chung-Ang University



2025.01.08

Outline

- ❖ **Introduction**
- ❖ **Problem Definition**
- ❖ **Findings**
 - ❑ *One-size-fits-all*
 - ❑ The Specialists
- ❖ **Conclusion**

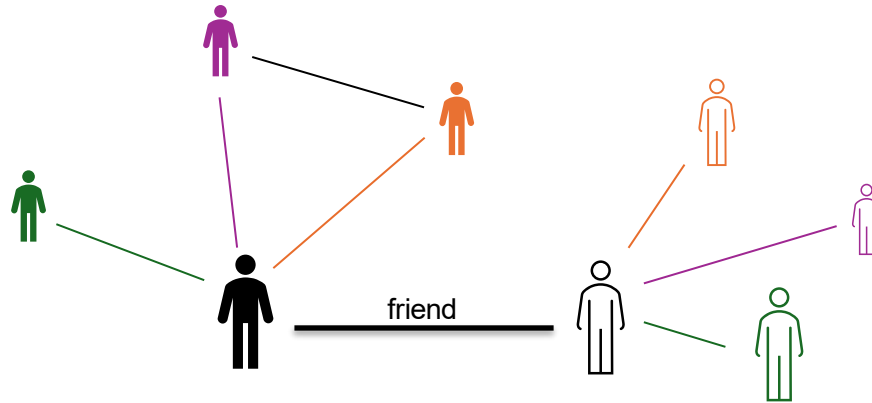
Introduction

Fairness-aware **Graph** Learning

Graph: Definition

❖ What is a Graph?

- ❑ A general language for describing and analyzing **entities** with **relations/interactions**
- ❑ General view: a collection of objects (i.e. nodes), along with a set of interactions (i.e. edges) between pairs of these object
- ❑ e.g., Social Network Graph
 - Nodes: Individuals
 - Edges: Friend(ship)



Graph: Definition

❖ Many types of data are graphs



Image credit: [Medium](#)
Social Networks

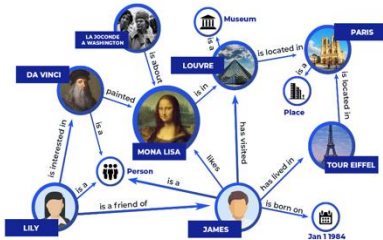


Image credit: [zilliz](#)
Knowledge Graph

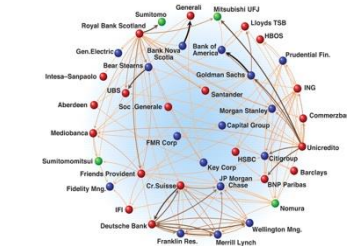


Image credit: [Science](#)
Economic Networks



Image credit: [missoulacurent](#)
Internet



Image credit: [Lumen Learning](#)
Communication Networks

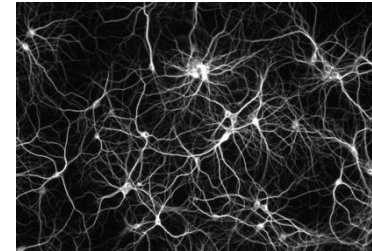
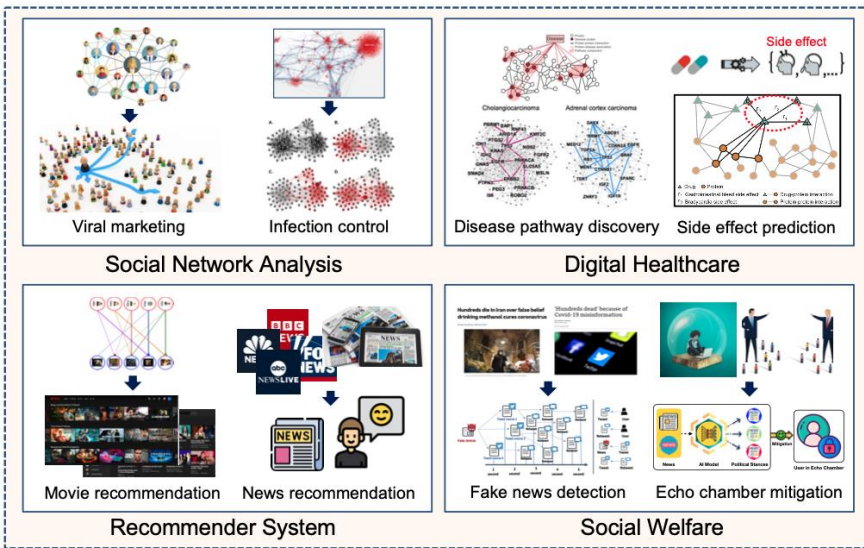
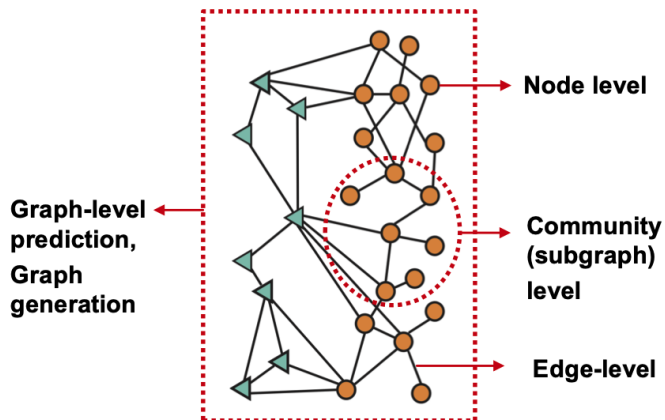


Image credit: [The Conversation](#)
Networks of Neurons

Graph: Application

❖ Different Types of Tasks

- ❑ Social Network Analysis
- ❑ Digital Healthcare
- ❑ Recommender System
- ❑ Social Welfare
- ❑ RAG, etc.



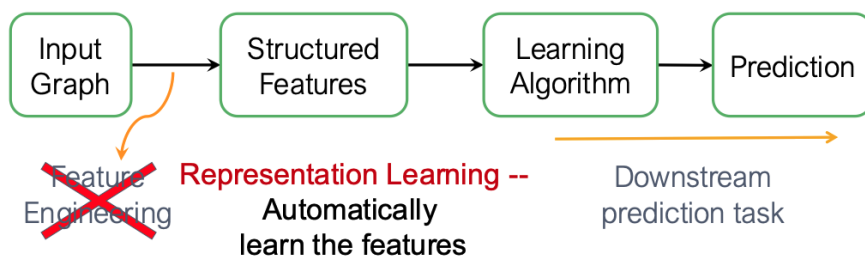
Introduction

Fairness-aware Graph **Learning**

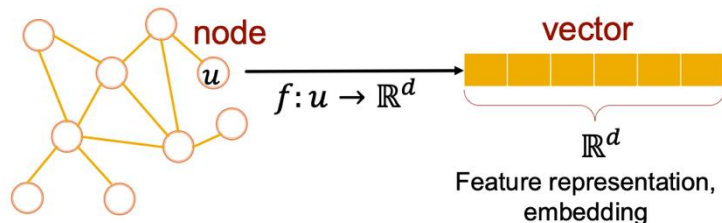
Graph Representation Learning

❖ Representation Learning on Graphs

- Given an input graph, then learn a model that maps features to labels
- No need to do feature engineering every single time



❖ Goal: Efficient task-independent feature Learning for machine learning with graphs



Tasks

- Node Classification
- Link Prediction
- Graph Classification
- Anomalous node detection
- Clustering
- ...

Graph Representation Learning

❖ Node Embedding

- ❑ Mappings into a low-dimensional space
- ❑ Protect similarities in network structure & nodal features
- ❑ Different approaches based on different similarity definitions

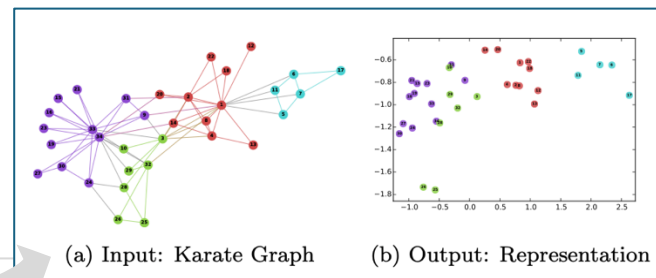
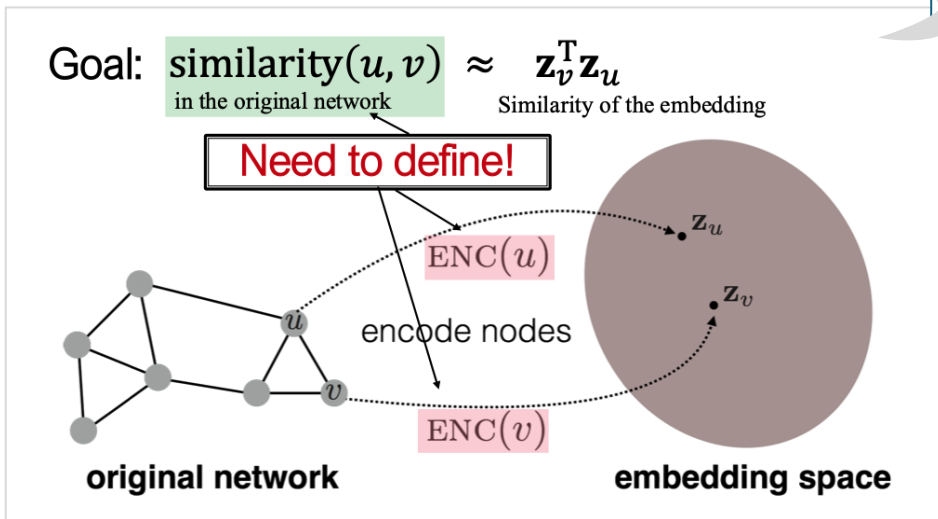


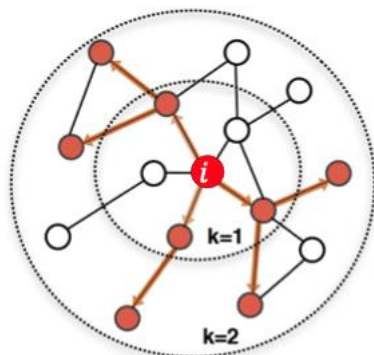
Image credit: [DeepWalk](#)

Here we will focus
on **Graph Neural
Networks!!**

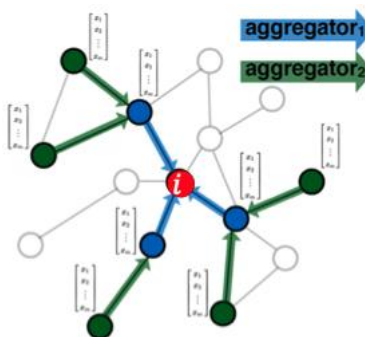
Graph Representation Learning

❖ Graph Neural Networks (GNNs)

- ❑ Widely adopted and versatile methods to handle predictive tasks on graphs
- ❑ Idea: Node's neighborhood defines a computation graph
- ❑ **Learn how to propagate information** across the graph to compute node features



Determine node
computation graph



Propagate and
transform information

[Kipf and Welling, ICLR 2017]

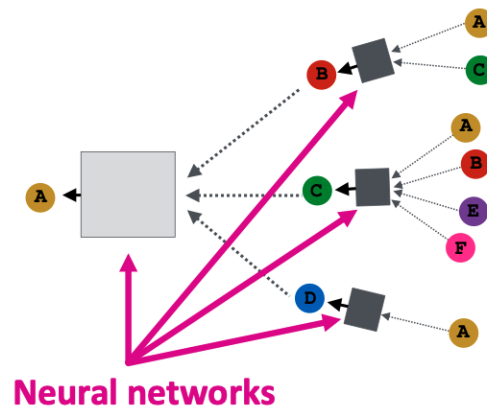
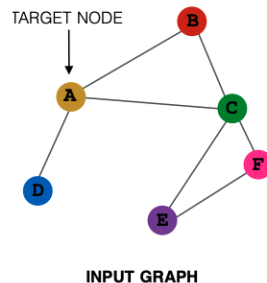
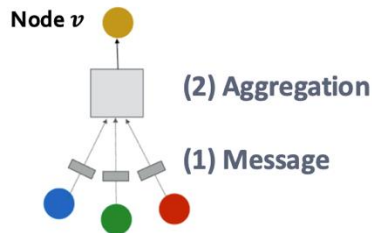
Graph Neural Networks

❖ Model Design: Overview

- Key Idea: Generate node embeddings based on local network neighborhoods
- Nodes aggregate information from their neighbors using neural networks
 - **(1) Message:** each node computes a message

$$\mathbf{m}_u^{(l)} = \text{MSG}^{(l)}(\mathbf{h}_u^{(l-1)}), u \in \{N(v) \cup v\}$$
 - **(2) Aggregation:** aggregate messages from neighbors

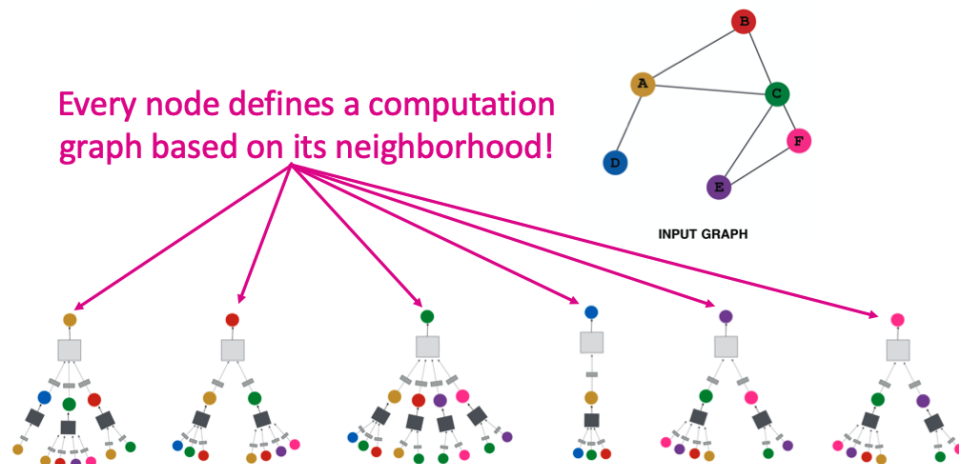
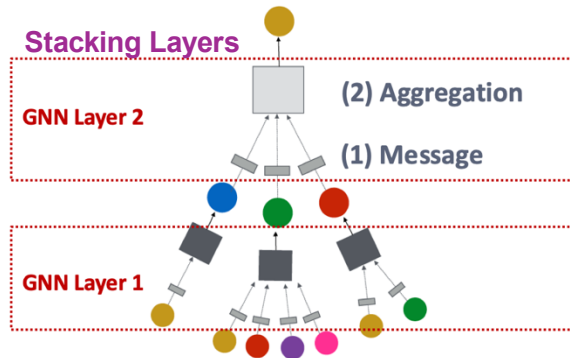
$$\mathbf{h}_v^{(l)} = \text{AGG}^{(l)}(\{\mathbf{m}_u^{(l)}, u \in N(v)\}, \mathbf{m}_v^{(l)})$$



Graph Neural Networks

❖ Model Design: Overview

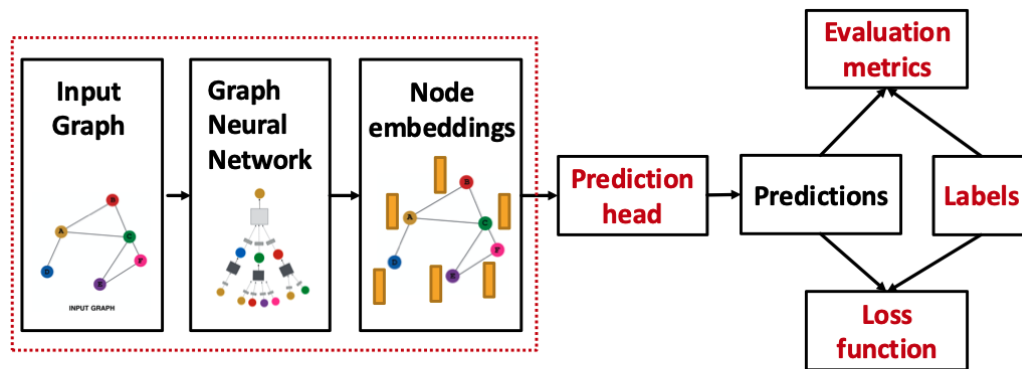
- ❑ Key Idea : Generate node embeddings based on local network neighborhoods
- ❑ Nodes aggregate information from their neighbors using neural networks
- ❑ Stacking more layers, a node can incorporate information from more hops
- ❑ Raw input graph $\neq$ Computational graph



Graph Neural Networks

❖ GNNs Training Pipeline

- ❑ **GNN acts as an encoder**, mapping the graph structure into a latent space of node embeddings
- ❑ Output of a GNN: a high-quality representation for every node in the graph
- ❑ These embeddings are fed into a **task-specific prediction head** and evaluated against real labels



Output of a GNN: set of node embeddings

$$\{\mathbf{h}_v^{(L)}, \forall v \in G\}$$

Application

❖ Credit Scoring

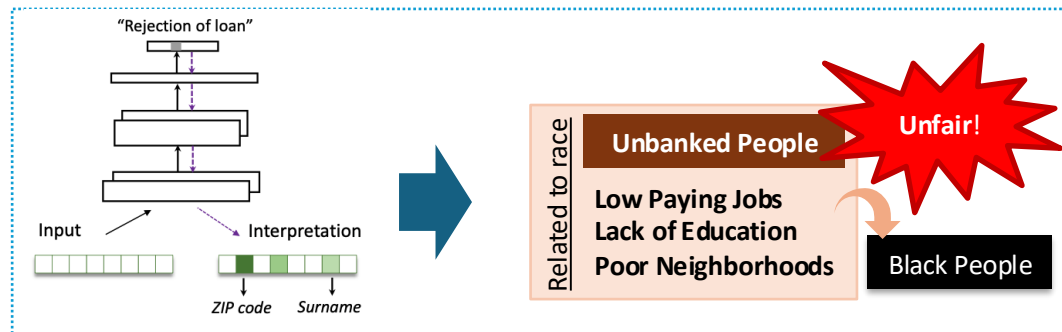
- ❑ Task: To decide whether to approve or deny loan applications based on applicants' information
- ❑ e.g., A credit scoring system, known as FICO, widely used in banking industry
- ❑ The FICO system is often viewed as objective regarding the decision-making process



Securing a loan has always been a challenging process

Application: Credit Scoring

❖ Securing a loan has always been a challenging process



- ❑ **Historical data** may include previous discriminatory decisions dominated by sensitive features
- ❑ Node representation learned from such data: **explicitly inherit the existing societal biases & exhibit unfairness**

Credit Scoring: Data Perspective

❖ Homophily Principle: Similarity breeds connection

- ❑ A contact between similar people occurs at a higher rate than among dissimilar people
- ❑ This principle structures network ties of every type
- ❑ People's personal networks are homogeneous with regard to behavioral, and intrapersonal characteristics
- ❑ Has powerful implications for the information they receive

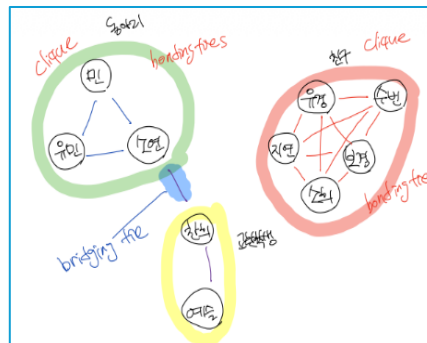
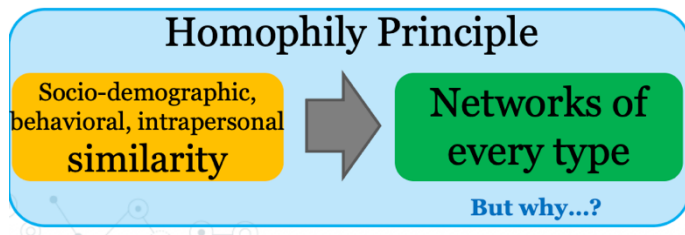


Image credit: Social Networks class @ CAU

▪ Information
▪ Attitudes
▪ Interactions

Social Divide
by race, age,
religion, and so on.

Consequences...?

Credit Scoring: Data Perspective

❖ Securing a loan has always been a challenging process

- ❑ **Network typology** also serves as an **implicit source** of societal bias
- ❑ **Homophily Effects:** Similarity breeds connection
- ❑ **Nodes with the same sensitive attribute tend to link with each other**

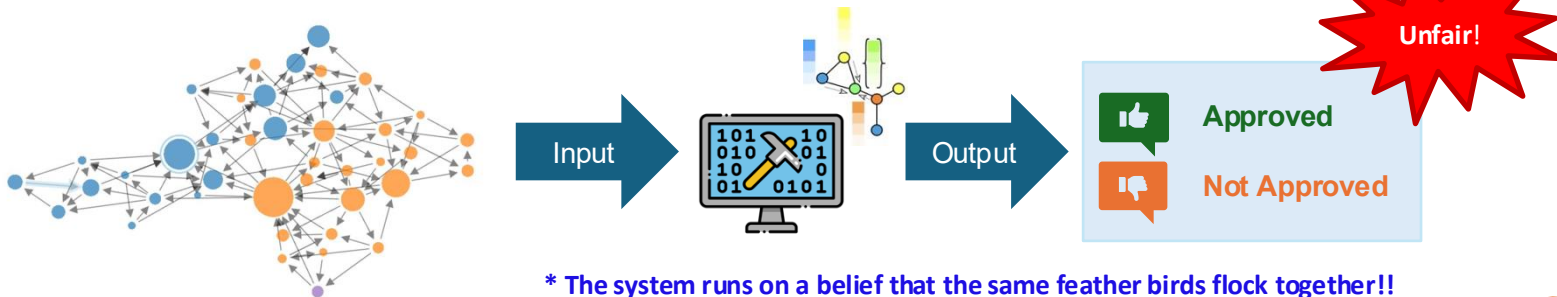
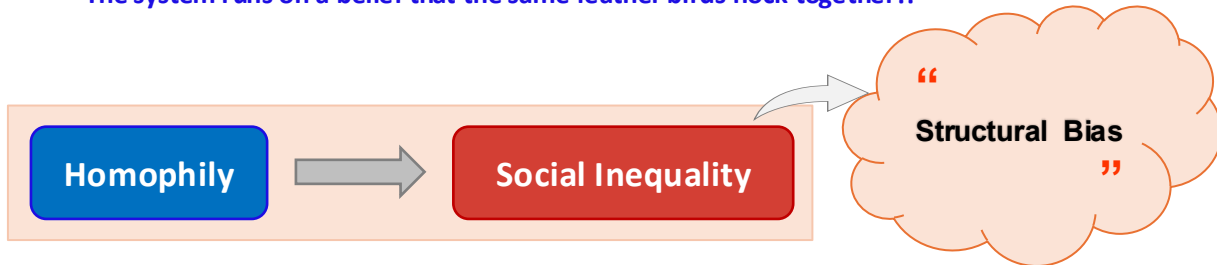


Image credit: bis.org

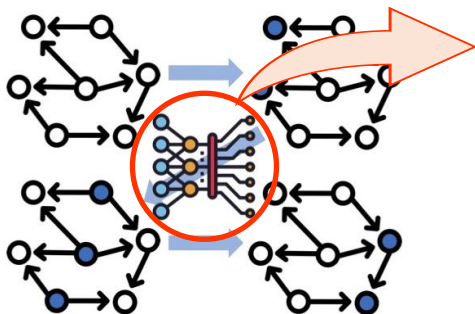


Credit Scoring: Model Perspective

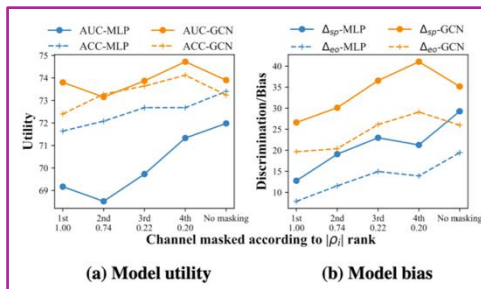
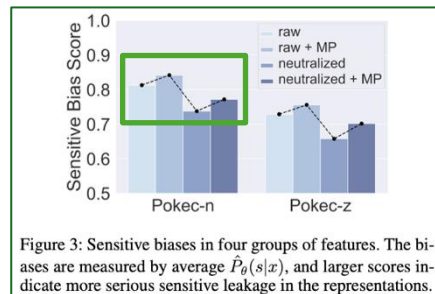
❖ Securing a loan has always been a challenging process

- ❑ **GNN's message passing mechanism exacerbates sensitive biases (attribute & structure biases)**
- ❑ Biases related to sensitive attributes can intentionally spread to non-sensitive attributes during GNN process

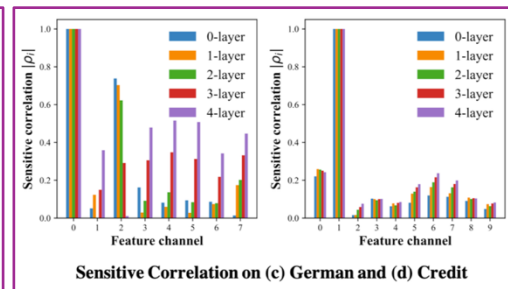
- (1) Sensitive information leakage
- (2) Feature propagation



Graph Neural Networks



(1) Sensitive Information Leakage



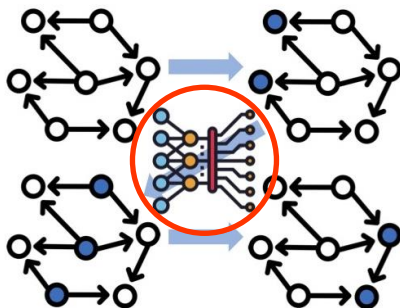
(2) Feature Propagation

Credit Scoring: Model Perspective

❖ Securing a loan has always been a challenging process

- ❑ **GNN's message passing mechanism exacerbates sensitive biases (attribute & structure biases)**
- ❑ Biases related to sensitive attributes can intentionally spread to non-sensitive attributes during GNN process

- **(1) Sensitive information leakage** and Correlation Variation
- **(2) Feature propagation**



Graph Neural Networks

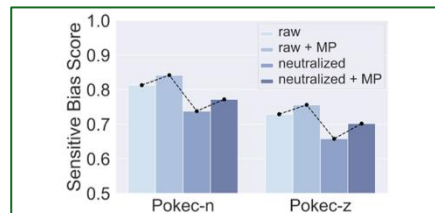
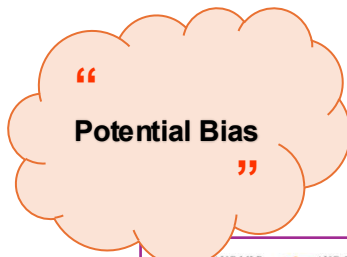
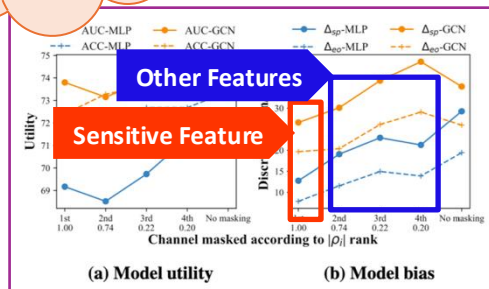


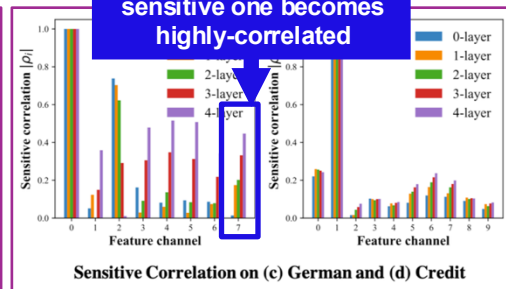
Figure 3: Sensitive biases in four groups of features. The biases are measured by average $\hat{P}_\theta(s|x)$, and indicate more serious sensitive leakage in the



(a) Model utility

(b) Model bias

(1) Sensitive Information Leakage



Sensitive Correlation on (c) German and (d) Credit

(2) Feature Propagation

Why Fairness is Considered in Graph Learning

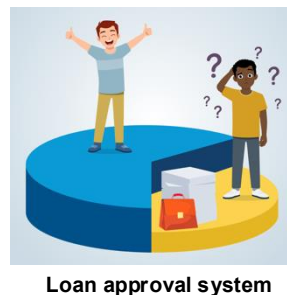
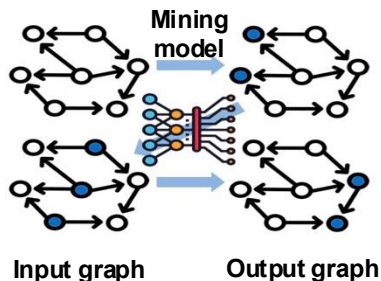
❖ Bias in nodal features will be encoded in node embeddings

- ❑ Aggregating information from neighbors → node embeddings

❖ Neighbors generally with same sensitive attributes

- ❑ Embeddings correlated with sensitive attributes
- ❑ Bias in graph structure propagated towards node embeddings

❖ Intertwined bias from both nodal features and graph structure



Addressing the **fairness concerns** for graph learning methods has become an urgent need, especially under **high-stake real-world applications**

Introduction

Fairness-aware Graph Learning

Fairness: Definition

❖ Principle

- ❑ Lack of favoritism from one side or another

❖ Various Definition of Fairness



Fairness definition

Two sides

Group fairness

Two demographic groups

Individual fairness

Two data points

Counterfactual fairness

A data point and its counterfactual version

Difference principle

Two groups of points with different utility



Images credit: [idiro](#)

Group Fairness: Definition

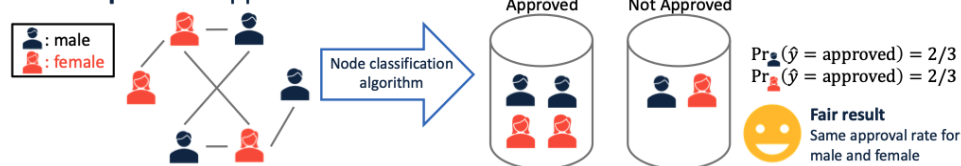
❖ The algorithm should **not** yield **discriminatory predictions or decisions against individuals from any specific sensitive subgroup**

❖ Fairness Metrics

□ Statistical Parity (a.k.a Demographic Parity) – SP/DP

➤ Equal acceptance rate

• Example: loan approval



□ Equal Opportunity - EO

➤ Equal true positive rate

• Example: loan approval



Images credit: [idiro](#)

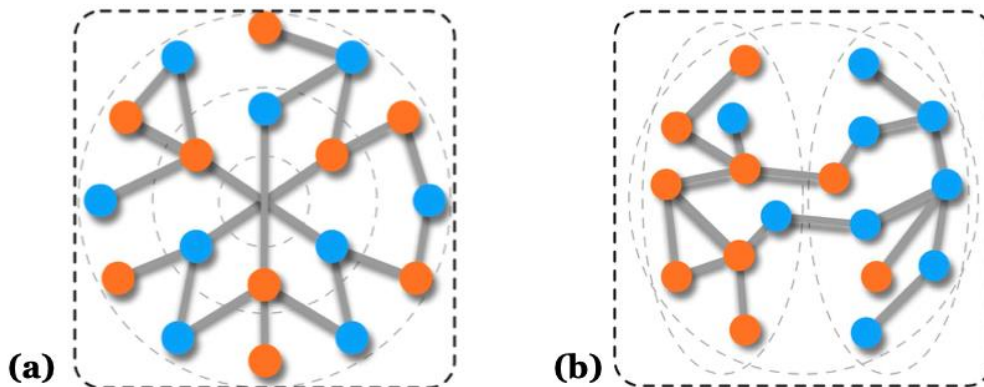
Problem Definition

What we want to solve?

Problem Definition

❖ Unique Challenges of Fulfilling Fairness in Graph ML Algorithms

- ❑ (1) Formulating proper fairness notions as the criteria to determine the existence of unfairness (i.e., bias)
 - **Fairness is not only about predictions, but about how relational information should (or should not) be used**
- ❑ (2) Preventing the graph ML algorithms from inheriting the bias exhibited in the input graphs

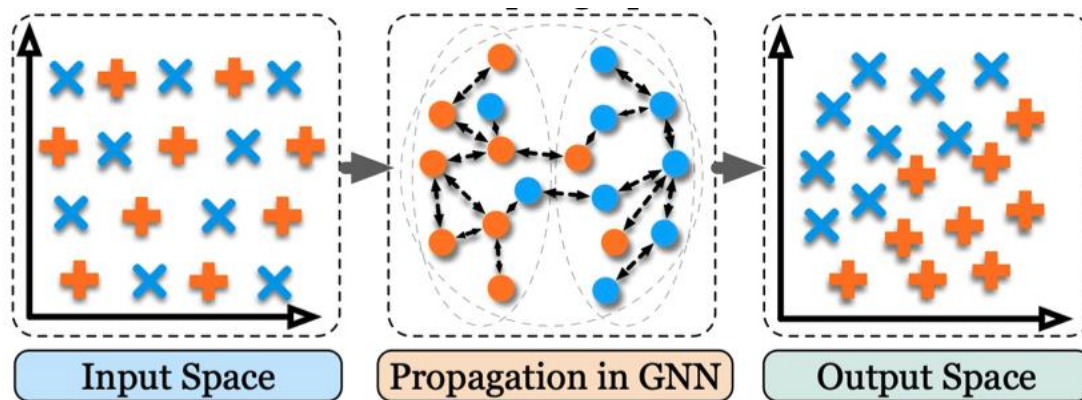


Compared with the structure in (a), the bias in the graph structure of (b) could lead to biased embeddings in Graph Neural Networks (GNNs).

Problem Definition

❖ Unique Challenges of Fulfilling Fairness in Graph ML Algorithms

- ❑ (1) Formulating proper fairness notions as the criteria to determine the existence of unfairness (i.e., bias)
- ❑ (2) Preventing the graph ML algorithms from inheriting the bias exhibited in the input graphs
 - Bias is not only learned—it is *propagated* and *amplified* by the graph itself

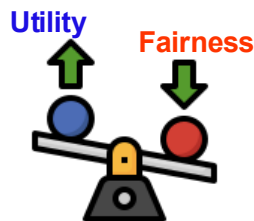
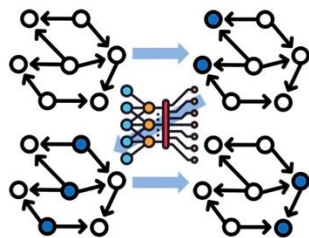


An example in Graph Neural Networks (GNNs): the unbalance between intra-group and inter-group edges could easily induce bias in the outcome space ^[1].

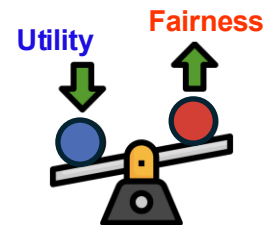
Problem Definition

❖ Goal of Fairness-aware Graph Learning

- ❑ Learn node embeddings that are both fair and accurate
- ❑ i.e. removing biases related to sensitive attribute while maintaining the accuracy of downstream tasks
- ❑ Accuracy and Fairness in a Trade-off Relationship



How can we achieve
both high *utility* and high *fairness*?



Findings

Fairness-aware Graph Learning Methods

One-size-fits-all

1. FairGNN

2. FairVGNN

3. FairSIN

The Specialists

4. EDITS

5. DAB-GNN

Poster Session I

WSDM '21, March 8–12, 2021, Virtual Event, Israel

Say No to the Discrimination: Learning Fair Graph Neural Networks with Limited Sensitive Attribute Information

Enyan Dai, Suhang Wang
The Pennsylvania State University
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FairGNN: Fairness with Limited Sensitive Attribute Information

❖ Motivation

- ❑ **Require abundant nodes** with known sensitive attributes ➡ either **revise the data** or **regularize the model**
- ❑ People are **unwilling to share their sensitive information** in the real world
 - Facebook: **only 14% of teen users public their complete profiles**
- ❑ Resulting in inadequate nodes with sensitive attributes known for fair model learning



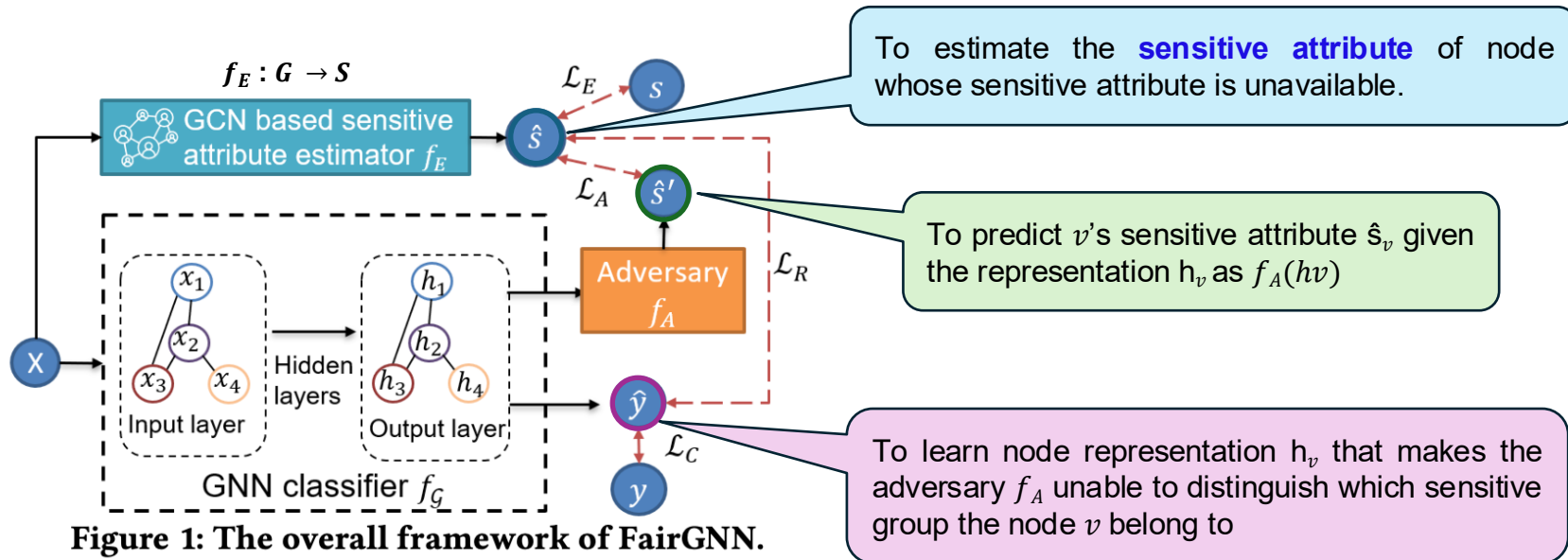
❖ Problem: Learning fair graph neural networks with limited sensitive information

- ❑ How to overcome the shortage of sensitive attributes for eliminating discrimination
- ❑ How to ensure the fairness of the GNN classifier

FairGNN: Framework

❖ Model Architecture

- ❑ Eliminate the bias of the final layer representations



$$\min_{\theta_G, \theta_E} \max_{\theta_A} \mathcal{L}_C + \mathcal{L}_E + \alpha \mathcal{L}_R - \beta \mathcal{L}_A$$

= Fair node representation

FairGNN: Experiment

❖ Observation

- FairGNN achieves comparable node classification accuracy with a much smaller bias (fairness)

Dataset	Metrics	GCN	GAT	ALFR	ALFR-e	Debias	Debias-e	FCGE	FairGCN	FairGAT
Pokec-z	ACC (%)	70.2 $\pm$ 0.1	70.4 $\pm$ 0.1	65.4 $\pm$ 0.3	68.0 $\pm$ 0.6	65.2 $\pm$ 0.7	67.5 $\pm$ 0.7	65.9 $\pm$ 0.2	70.0 $\pm$ 0.3	70.1 $\pm$ 0.1
	AUC (%)	77.2 $\pm$ 0.1	76.7 $\pm$ 0.1	71.3 $\pm$ 0.3	74.0 $\pm$ 0.7	71.4 $\pm$ 0.6	74.2 $\pm$ 0.7	71.0 $\pm$ 0.2	76.7 $\pm$ 0.2	76.5 $\pm$ 0.2
	Δ_{SP} (%)	9.9 $\pm$ 1.1	9.1 $\pm$ 0.9	2.8 $\pm$ 0.5	5.8 $\pm$ 0.4	1.9 $\pm$ 0.6	4.7 $\pm$ 1.0	3.1 $\pm$ 0.5	0.9 $\pm$ 0.5	0.5 $\pm$ 0.3
	Δ_{EO} (%)	9.1 $\pm$ 0.6	8.4 $\pm$ 0.6	1.1 $\pm$ 0.4	2.8 $\pm$ 0.8	1.9 $\pm$ 0.4	3.0 $\pm$ 1.4	1.7 $\pm$ 0.6	1.7 $\pm$ 0.2	0.8 $\pm$ 0.3
Pokec-n	ACC (%)	70.5 $\pm$ 0.2	70.3 $\pm$ 0.1	63.1 $\pm$ 0.6	66.2 $\pm$ 0.5	62.6 $\pm$ 0.9	65.6 $\pm$ 0.8	64.8 $\pm$ 0.5	70.1 $\pm$ 0.2	70.0 $\pm$ 0.2
	AUC (%)	75.1 $\pm$ 0.2	75.1 $\pm$ 0.2	67.7 $\pm$ 0.5	71.9 $\pm$ 0.3	67.9 $\pm$ 0.7	71.7 $\pm$ 0.7	69.5 $\pm$ 0.4	74.9 $\pm$ 0.4	74.9 $\pm$ 0.4
	Δ_{SP} (%)	9.6 $\pm$ 0.9	9.4 $\pm$ 0.7	3.05 $\pm$ 0.5	4.1 $\pm$ 0.5	2.4 $\pm$ 0.7	3.6 $\pm$ 0.2	4.1 $\pm$ 0.8	0.8 $\pm$ 0.2	0.6 $\pm$ 0.3
	Δ_{EO} (%)	12.8 $\pm$ 1.3	12.0 $\pm$ 1.5	3.9 $\pm$ 0.6	4.6 $\pm$ 1.6	2.6 $\pm$ 1.0	4.4 $\pm$ 1.2	5.5 $\pm$ 0.9	1.1 $\pm$ 0.5	0.8 $\pm$ 0.2
NBA	ACC (%)	71.2 $\pm$ 0.5	71.9 $\pm$ 1.1	64.3 $\pm$ 1.3	66.0 $\pm$ 0.4	63.1 $\pm$ 1.1	65.6 $\pm$ 2.4	66.0 $\pm$ 1.5	71.1 $\pm$ 1.0	71.5 $\pm$ 0.8
	AUC (%)	78.3 $\pm$ 0.3	78.2 $\pm$ 0.6	71.5 $\pm$ 0.3	72.9 $\pm$ 1.0	71.3 $\pm$ 0.7	72.9 $\pm$ 1.2	73.6 $\pm$ 1.5	77.0 $\pm$ 0.3	77.5 $\pm$ 0.7
	Δ_{SP} (%)	7.9 $\pm$ 1.3	10.2 $\pm$ 2.5	2.3 $\pm$ 0.9	4.7 $\pm$ 1.8	2.5 $\pm$ 1.5	5.3 $\pm$ 0.9	2.9 $\pm$ 1.0	1.0 $\pm$ 0.5	0.7 $\pm$ 0.5
	Δ_{EO} (%)	17.8 $\pm$ 2.6	15.9 $\pm$ 4.0	3.2 $\pm$ 1.5	4.7 $\pm$ 1.7	3.1 $\pm$ 1.9	3.1 $\pm$ 1.3	3.0 $\pm$ 1.2	1.2 $\pm$ 0.4	0.7 $\pm$ 0.3

Paper \ Problems	Attribute Bias	Structural Bias	Potential Bias
FairGNN	0	X	Δ



Improving Fairness in Graph Neural Networks via Mitigating Sensitive Attribute Leakage

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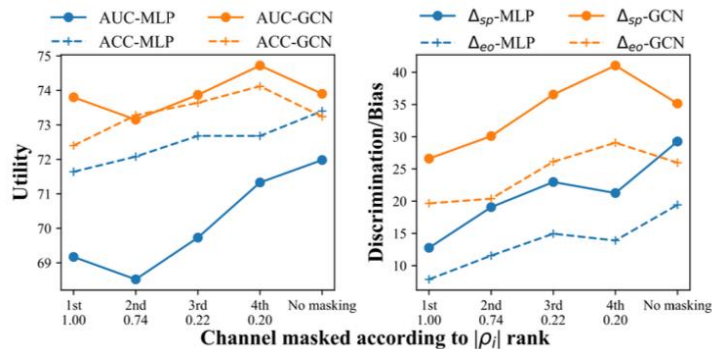
Tyler Derr

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FairVGNN: Fair View Graph Neural Network

❖ Problem & Observations I

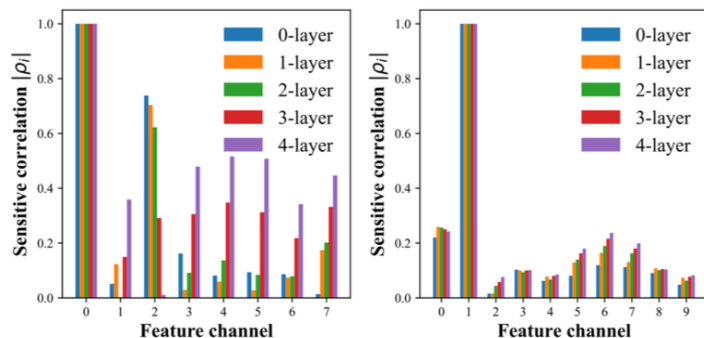
- ❑ Feature Propagation can vary feature correlations
- ❑ Cause **sensitive information leakage** to innocuous feature channels
- ❑ Could further exacerbate discrimination in predictions



(a) Model utility

(b) Model bias

Leakage of Sensitive Information
to other **non-sensitive** feature channels



Sensitive Correlation on (c) German and (d) Credit

Correlations between the sensitive features and other channels change after **Feature Propagation in GNNs**

FairVGNN: Fair View Graph Neural Networks

❖ Problem & Observations II

- ❑ S2 achieves lower Δ_{SP} , Δ_{EO} than S1 for both MLP and GCN on German
 - The rank of sensitive correlation changes after feature propagation and masking according to S2 leads to better fairness
 - Importance of considering feature propagation in determining which feature channels are more sensitive-correlated and required to be masked

Table 1: Evaluating model utility and fairness when using

- Automatically learn **fair views** by identifying and masking sensitive-correlated channels
- Adaptively **clamping weights** to avoid leveraging sensitive-related features in learning fair node representations

GIN	S ₀	72.71	82.78	13.56	9.47	74.36	82.28	14.48	12.35
	S ₁	71.66	82.50	3.01	1.72	73.44	83.23	14.29	11.79
	S ₂	70.77	83.53	1.46	2.67	73.28	83.27	13.96	11.34

* S₀: training using the original feature matrix **X** without any masking.

* S₁/S₂: training with masking the top-4 channels based on the rank of $\rho^{\text{origin}}/\rho^{\text{prop}}$.

FairVGNN: Framework

❖ Architecture

❑ Fair Feature View Generation

$$\tilde{\mathbf{X}} = \mathbf{X} \odot \mathbf{m} = [\mathbf{X}_1^\top \odot \mathbf{m}, \mathbf{X}_2^\top \odot \mathbf{m}, \dots, \mathbf{X}_n^\top \odot \mathbf{m}],$$

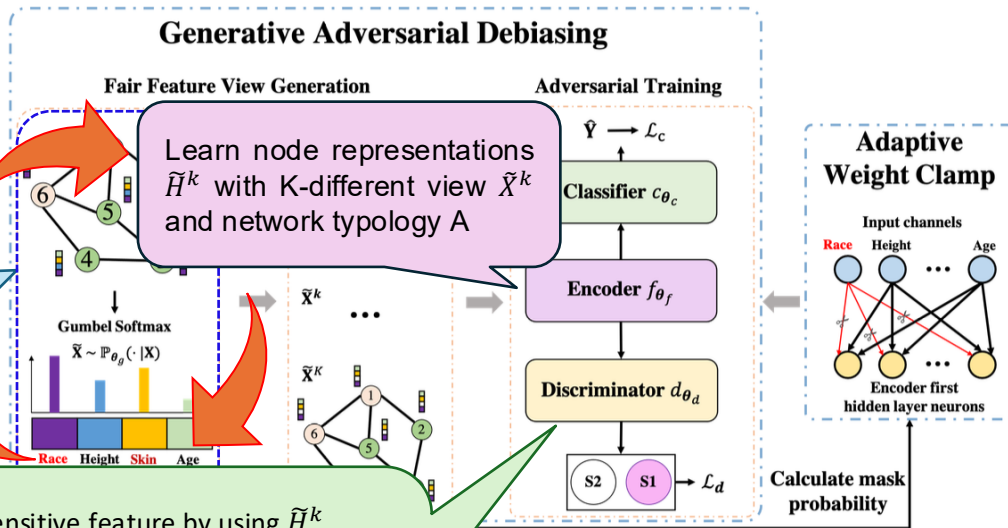
A learnable latent distribution:

- Sample K-different masks and generate K-corresponding views
- Be updated towards generating **less-biased views** $\tilde{\mathbf{X}}$
- Be fed to the encoder

❑ Adaptive Weight Clamping

$$\mathbf{W}_{ij}^{f,1} = \begin{cases} \mathbf{W}_{ij}^{f,1}, \\ \text{sign}(\mathbf{W}_{ij}^{f,1}) * \epsilon * \mathbf{p}_j, \end{cases}$$

- Predict nodes' sensitive feature by using $\tilde{\mathbf{H}}^k$
- Pave a way to obtain the **optimal fair view** generator: **encode** as much **task-relevant** information while **discarding** as much **bias-relevant** information as possible



ow of the Fair View Graph Neural Net-
th two main modules: (a) generative ad-
learn fair view of features and (b) adap-
to clamp weights of sensitive-related
der.

FairVGNN: Experiment

❖ Observation

- Consistently perform the best compared with other bias-alleviating methods

(In terms of the average rank for all datasets and across all evaluation metrics)

Table 3: Model utility and bias of node classification. We compare the proposed FairVGNN (i.e., FairV) against state-of-the-art baselines NIFTY, EDITS, and FairGNN (i.e., Fair) when equipped with various GNN backbones (i.e., GCN, GIN, and SAGE). The best and runner-up results are colored in **red and **blue**. $\uparrow$ represents the larger, the better while $\downarrow$ represents the opposite.**

Encoder	Method	German					Credit					Bail					Avg. (Rank)
		AUC ($\uparrow$)	F1 ($\uparrow$)	ACC ($\uparrow$)	Δ_{sp} ($\downarrow$)	Δ_{eo} ($\downarrow$)	AUC ($\uparrow$)	F1 ($\uparrow$)	ACC ($\uparrow$)	Δ_{sp} ($\downarrow$)	Δ_{eo} ($\downarrow$)	AUC ($\uparrow$)	F1 ($\uparrow$)	ACC ($\uparrow$)	Δ_{sp} ($\downarrow$)	Δ_{eo} ($\downarrow$)	
GCN	Vanilla	74.11$\pm$0.37	82.46 $\pm$ 0.89	73.44$\pm$1.09	35.17 $\pm$ 7.27	25.17 $\pm$ 5.89	73.87 $\pm$ 0.02	81.92 $\pm$ 0.02	73.67 $\pm$ 0.03	12.86 $\pm$ 0.09	10.63 $\pm$ 0.13	87.08 $\pm$ 0.35	79.02 $\pm$ 0.74	84.56 $\pm$ 0.68	7.35 $\pm$ 0.72	4.96 $\pm$ 0.62	9.17
	NIFTY	68.78 $\pm$ 2.69	81.40 $\pm$ 0.54	69.92 $\pm$ 1.14	5.73 $\pm$ 5.25	5.08 $\pm$ 4.29	71.96 $\pm$ 0.19	81.72 $\pm$ 0.05	73.45 $\pm$ 0.06	11.68 $\pm$ 0.07	9.39 $\pm$ 0.07	78.20 $\pm$ 2.78	64.76 $\pm$ 3.91	74.19 $\pm$ 2.57	2.44 $\pm$ 1.29	1.72 $\pm$ 1.08	9.69
	EDITS	69.41 $\pm$ 2.33	81.55 $\pm$ 0.59	71.60 $\pm$ 0.89	4.05 $\pm$ 4.48	3.89 $\pm$ 4.23	73.01 $\pm$ 0.11	81.81 $\pm$ 0.28	73.51 $\pm$ 0.30	10.90 $\pm$ 1.22	8.75 $\pm$ 1.21	86.44 $\pm$ 2.17	75.58 $\pm$ 3.77	84.49 $\pm$ 2.27	6.64 $\pm$ 0.39	7.51 $\pm$ 1.20	9.89
	FairGNN	67.35 $\pm$ 2.13	82.01 $\pm$ 0.26	69.68 $\pm$ 0.30	3.49 $\pm$ 2.15	3.40 $\pm$ 2.15	71.95 $\pm$ 1.43	81.84 $\pm$ 1.19	73.41 $\pm$ 1.24	12.64 $\pm$ 2.11	10.41 $\pm$ 2.03	87.36 $\pm$ 0.90	77.50 $\pm$ 1.69	82.94 $\pm$ 1.67	6.90 $\pm$ 0.17	4.65 $\pm$ 0.14	9.17
	FairVGNN	72.41 $\pm$ 2.10	82.14 $\pm$ 0.42	70.16 $\pm$ 0.86	1.71 $\pm$ 1.68	0.88$\pm$0.58	71.34 $\pm$ 0.41	87.08 $\pm$ 0.74	78.04 $\pm$ 0.33	5.02 $\pm$ 5.22	3.60 $\pm$ 4.31	85.68 $\pm$ 0.37	79.11 $\pm$ 0.33	84.73 $\pm$ 0.46	6.53 $\pm$ 0.67	4.95 $\pm$ 1.22	5.67
GIN	Vanilla	72.71 $\pm$ 1.44	82.78$\pm$0.50	73.84$\pm$0.54	13.56 $\pm$ 5.23	9.47 $\pm$ 4.49	74.36 $\pm$ 0.21	82.28 $\pm$ 0.64	74.02 $\pm$ 0.73	14.48 $\pm$ 2.44	12.35 $\pm$ 2.86	86.14 $\pm$ 0.25	76.49 $\pm$ 0.57	81.70 $\pm$ 0.67	8.55 $\pm$ 1.61	6.99 $\pm$ 1.51	9.56
	NIFTY	67.61 $\pm$ 4.88	80.46 $\pm$ 3.06	69.92 $\pm$ 3.64	5.26 $\pm$ 3.24	5.34 $\pm$ 5.67	70.90 $\pm$ 0.24	84.05 $\pm$ 0.82	75.59 $\pm$ 0.66	7.09 $\pm$ 4.62	6.22 $\pm$ 3.26	82.33 $\pm$ 4.61	70.64 $\pm$ 6.73	74.46 $\pm$ 9.98	5.57 $\pm$ 1.11	3.41 $\pm$ 1.43	8.56
	EDITS	69.35 $\pm$ 1.64	82.80 $\pm$ 0.22	72.08 $\pm$ 0.66	0.86 $\pm$ 0.76	1.72 $\pm$ 1.14	72.35 $\pm$ 1.11	82.47 $\pm$ 0.85	74.07 $\pm$ 0.98	14.11 $\pm$ 14.45	15.40 $\pm$ 15.76	80.19 $\pm$ 4.62	68.07 $\pm$ 5.30	73.74 $\pm$ 5.12	6.71 $\pm$ 2.35	5.98 $\pm$ 3.66	11.36
	FairGNN	72.95 $\pm$ 0.82	83.16$\pm$0.56	72.24 $\pm$ 1.44	6.88 $\pm$ 4.42	2.06 $\pm$ 1.46	68.66 $\pm$ 4.48	79.47 $\pm$ 5.29	70.33 $\pm$ 5.50	4.67$\pm$3.06	3.94 $\pm$ 1.49	86.14 $\pm$ 0.89	73.67 $\pm$ 1.17	77.90 $\pm$ 2.21	6.33 $\pm$ 1.49	4.74 $\pm$ 1.64	7.64
	FairVGNN	71.65 $\pm$ 1.90	82.40 $\pm$ 0.14	70.16 $\pm$ 0.32	0.43$\pm$0.54	0.34$\pm$0.41	71.36 $\pm$ 0.72	87.44$\pm$0.23	78.18$\pm$0.30	2.85$\pm$2.01	1.72$\pm$1.80	83.22 $\pm$ 1.60	76.36 $\pm$ 2.20	83.86 $\pm$ 1.57	5.67 $\pm$ 0.76	5.77 $\pm$ 1.26	5.44
SAGE	Vanilla	75.74$\pm$0.69	81.25 $\pm$ 1.72	72.24 $\pm$ 1.61	24.30 $\pm$ 6.93	15.55 $\pm$ 7.59	74.58$\pm$1.31	83.38 $\pm$ 0.77	75.28 $\pm$ 0.83	15.65 $\pm$ 1.30	13.34 $\pm$ 1.34	90.71 $\pm$ 0.69	80.99 $\pm$ 0.55	86.72 $\pm$ 0.48	2.16 $\pm$ 1.53	0.84$\pm$0.55	7.31
	NIFTY	72.05 $\pm$ 2.15	79.20 $\pm$ 1.19	69.60 $\pm$ 1.50	7.74 $\pm$ 7.80	5.17 $\pm$ 2.38	72.89 $\pm$ 0.44	82.60 $\pm$ 1.25	74.39 $\pm$ 1.35	10.65 $\pm$ 1.65	8.10 $\pm$ 1.91	92.04$\pm$0.89	77.81 $\pm$ 6.03	84.11 $\pm$ 5.49	5.74 $\pm$ 0.38	4.07 $\pm$ 1.28	8.06
	EDITS	69.76 $\pm$ 5.46	81.04 $\pm$ 1.09	71.68 $\pm$ 1.25	8.42 $\pm$ 7.35	5.69 $\pm$ 2.16	75.04 $\pm$ 0.12	82.41 $\pm$ 0.52	74.13 $\pm$ 0.59	11.34 $\pm$ 6.36	9.38 $\pm$ 5.39	89.07 $\pm$ 2.26	77.83 $\pm$ 3.79	84.42 $\pm$ 2.87	3.74 $\pm$ 3.54	4.46 $\pm$ 3.50	11.36
	FairGNN	65.85 $\pm$ 9.49	82.29 $\pm$ 0.32	70.64 $\pm$ 0.74	7.65 $\pm$ 8.07	4.18 $\pm$ 4.86	70.82 $\pm$ 0.74	83.97 $\pm$ 2.00	75.29 $\pm$ 1.62	6.17 $\pm$ 5.57	5.06 $\pm$ 4.46	91.53 $\pm$ 0.38	82.55$\pm$0.98	87.68$\pm$0.73	1.94$\pm$0.82	1.72 $\pm$ 0.70	5.83
	FairVGNN	73.84 $\pm$ 0.52	81.91 $\pm$ 0.63	70.00 $\pm$ 0.25	1.36$\pm$1.90	1.22 $\pm$ 1.49	74.05$\pm$0.20	87.84$\pm$0.32	79.94$\pm$0.30	4.94 $\pm$ 1.10	2.39$\pm$0.71	91.56$\pm$1.71	83.58$\pm$1.88	88.41$\pm$1.29	1.14$\pm$0.67	1.69$\pm$1.13	2.92

Paper \ Problems	Attribute Bias	Structural Bias	Potential Bias
	0	X	Δ
FairVGNN	0	X	Δ



FairSIN: Achieving Fairness in Graph Neural Networks through Sensitive Information Neutralization

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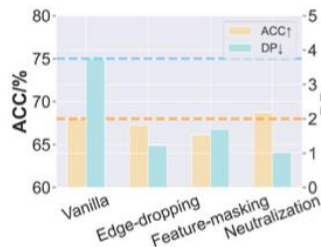
FairSIN: Fair GNN through Sensitive Information Neutralization

❖ Motivation

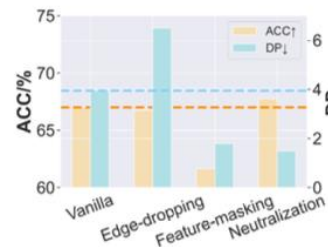
- Recent SOTA methods: Filtering-based methods
- Filter out sensitive from inputs or representations (e.g., feature masking or edge dropping)
- Filter also out non-sensitive feature information → sub-optimal trade-off between utility and fairness



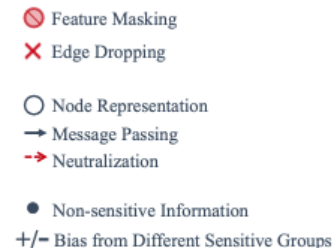
(a) Original Message Passing (b) Filtering-Based Method



(a) Pokcec-n



(b) Pokcec-z



Motivation verification on Pokcec datasets

What about **bias-neutralization**:

adding *theoretically motivated corrective features* to cancel sensitive information without altering the dataset?

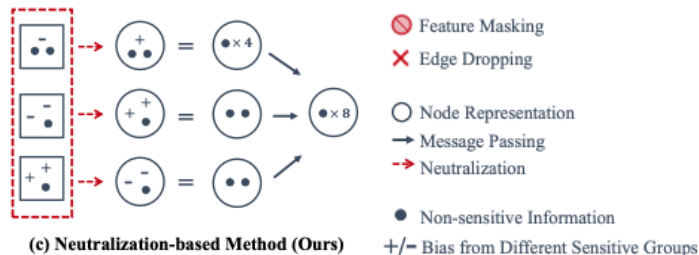
FairSIN: Fair GNN through Sensitive Information Neutralization

❖ A neutralization-based paradigm

- Introduce extra Fairness-facilitating Features (F3) to node features or representations
- Node features or representations can be debiased before message passing

❖ Real World Dataset with very few or even no heterogeneous neighbors

- Propose to train an estimator
- Predict the average features or representations of a node's heterogeneous neighbors



Dataset	Bail	Pokec-n	Pokec-z
# Nodes	18,876	66,569	67,797
# Features	18	266	277
# Edges	321,308	729,129	882,765
Node label	Bail decision	Working field	Working field
Sens. Attri.	Race	Region	Region
Avg. Deg.	34.04	16.53	19.23
Avg. H-Deg.	15.79	0.73	0.90

Table 1: Dataset statistics. “H-” means “heterogeneous”.

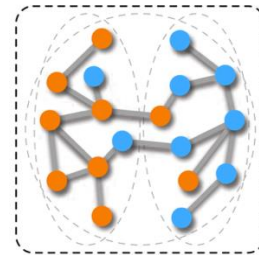
FairSIN: Framework

❖ Data-centric Variant

❑ (1) Graph modification (FairSIN-G):

- $\delta > 0$, hyper-parameter

$$\mathbf{A}_{ij} = \begin{cases} 1 + \delta, & \text{if } (v_i, v_j) \in \mathcal{E} \text{ and } s_i \neq s_j \\ 1, & \text{if } (v_i, v_j) \in \mathcal{E} \text{ and } s_i = s_j \\ 0, & \text{if } (v_i, v_j) \notin \mathcal{E} \end{cases}$$



(b) Biased graph topology

❑ (2) Feature modification (FairSIN-F)

- Compute average feature of each node v_i 's heterogeneous neighbors as $\mathbf{x}_i^{diff} = \frac{1}{|\mathcal{N}_i^{diff}|} \sum_{v_j \in \mathcal{N}_i^{diff}} \mathbf{x}_j$
- In real world graph, **very few or no heterogeneous neighbors**
 - Make calculation infeasible or very uncertain
- Train MLP to **estimate** $\mathbf{x}_i^{diff}$ using MSE loss: $\mathcal{L}_F = \frac{1}{|\mathcal{V}|} \sum_{i: |\mathcal{N}_i^{diff}| \geq 1} \|\text{MLP}_\phi(\mathbf{x}_i) - \mathbf{x}_i^{diff}\|^2$
- Neutralize each node feature as $\tilde{\mathbf{x}}_i = \mathbf{x}_i + \delta \text{MLP}_\phi(\mathbf{x}_i)$



Estimator

Nodes with rich heterogeneous neighbors can transfer their knowledge to other nodes

FairSIN: Framework

❖ Model-centric Variant

- Extend FairSIN-F by jointly learning the MLP_{ϕ} and GNN encoder

- MSE loss: $\min \min \mathcal{L}_F^k$

- $\delta = (0,1]$

$$\begin{aligned}\tilde{\mathbf{H}}^k &= \mathbf{H}^k + \delta^k \text{MLP}_{\phi}^k(\mathbf{H}^k), \\ \mathbf{H}^{k+1} &= \text{MessagePassing}(\tilde{\mathbf{H}}^k),\end{aligned}$$

- Introduce a **discriminator** module MLP_{ψ}

- Impose extra fairness constraints on the encoded representations

- Another MLP to implement the discriminator

- Predict the sensitive attribute based on the encoded(final) representations by GNN



Nodes with rich heterogeneous neighbors can provide additional information when calculating representations

FairSIN: Experiment

❖ Observation

- ❑ Neutralized-based strategy achieve a better trade-off than SOTA
 - Pokec-n/-z : very few hetero neighbors
- ❑ Both the best overall classification performance and group fairness under different GNN encoders
- ❑ Both FairSIN-G and FairSIN-F maintain the accuracy and improve the fairness on average

Encoder	Method	Bail			Pokec.n			Pokec.z		
		ACC↑	DP↓	EO↓	ACC↑	DP↓	EO↓	ACC↑	DP↓	EO↓
GCN	vanilla	87.55±0.54	6.85±0.47	5.26±0.78	<u>68.55±0.51</u>	3.75±0.94	2.93±1.15	<u>66.78±1.09</u>	3.95±1.03	2.76±0.95
	FairGNN	82.94±1.67	6.90±0.17	4.65±0.14	67.36±2.06	3.29±2.95	2.46±2.64	67.65±1.65	1.87±1.95	1.32±1.42
	EDITS	84.49±2.27	6.64±0.39	7.51±1.20	OOM	OOM	OOM	OOM	OOM	OOM
	NIFTY	82.36±3.91	5.78±1.29	4.72±1.08	67.24±0.49	1.22±0.94	2.79±1.24	66.74±0.93	6.50±2.16	7.64±1.77
	FairVGNN	84.73±0.46	6.53±0.67	4.95±1.22	66.10±1.45	1.69±0.79	1.78±0.70	61.64±4.72	1.79±1.22	1.25±1.01
	FairSIN-G	85.57±1.08	6.57±0.29	5.55±0.84	68.22±0.39	2.56±0.60	1.69±1.29	65.73±1.76	3.53±1.20	2.42±1.43
	FairSIN-F	87.61±0.83	5.54±0.40	3.47±1.03	67.96±1.54	1.16±0.90	0.98±0.70	66.38±1.39	2.53±0.97	2.03±1.23
	FairSIN w/o N.	87.26±0.17	5.93±0.04	4.30±0.20	68.35±0.62	2.51±1.99	2.36±1.89	65.87±1.34	1.98±1.01	1.87±0.64
	FairSIN w/o D.	87.40±0.15	5.65±0.40	4.63±0.52	<u>68.74±0.33</u>	2.22±1.47	1.67±1.70	66.42±1.52	2.73±1.08	2.37±0.69
	FairSIN	87.67±0.26	4.56±0.75	2.79±0.89	69.34±0.32	0.57±0.19	0.43±0.41	67.76±0.71	1.49±0.74	0.59±0.50
GIN	vanilla	83.52±0.87	7.55±0.51	6.17±0.69	69.25±1.75	3.71±1.20	2.55±1.52	65.83±1.31	1.97±1.12	2.17±0.48
	FairGNN	77.90±2.21	6.33±1.49	4.74±1.64	67.10±3.25	3.82±2.44	3.62±2.78	<u>66.49±1.54</u>	3.53±3.90	3.17±3.52
	EDITS	73.74±5.12	6.71±2.35	5.98±3.66	OOM	OOM	OOM	OOM	OOM	OOM
	NIFTY	74.46±9.98	5.57±1.11	3.41±1.43	66.37±1.51	3.84±1.05	3.24±1.60	65.57±1.34	2.70±1.28	3.23±1.92
	FairVGNN	83.86±1.57	5.67±0.76	5.77±0.76	68.37±0.97	1.88±0.99	1.24±1.06	65.46±1.22	1.45±1.13	1.21±1.06
	FairSIN-G	86.10±1.39	6.93±0.16	6.75±0.66	67.73±1.67	1.98±1.54	1.50±1.15	65.09±2.69	1.55±1.23	1.74±0.80
	FairSIN-F	<u>86.48±0.75</u>	5.95±1.85	5.97±2.07	68.92±1.08	<u>1.51±1.11</u>	0.82±0.79	65.97±0.82	1.45±1.15	1.14±0.73
	FairSIN w/o N.	85.27±0.70	7.21±0.39	6.75±0.55	68.92±1.13	2.81±1.91	2.12±1.30	65.04±1.56	2.19±1.96	1.23±0.92
	FairSIN w/o D.	86.44±0.80	<u>4.38±1.48</u>	4.23±1.88	70.04±0.80	2.44±1.50	1.63±1.24	65.58±0.71	<u>1.40±0.67</u>	<u>1.12±0.24</u>
	FairSIN	86.52±0.48	4.35±0.71	4.17±0.96	<u>69.58±0.57</u>	1.11±0.31	<u>0.97±0.59</u>	66.74±1.56	0.64±0.47	1.01±0.64
SAGE	vanilla	88.13±1.12	1.13±0.48	2.61±1.16	69.03±0.77	3.09±1.29	2.21±1.60	66.55±0.69	4.71±1.05	2.72±0.85
	FairGNN	87.68±0.73	1.94±0.82	1.72±0.70	67.03±2.61	2.97±1.28	2.06±3.02	<u>67.68±1.49</u>	<u>2.86±1.39</u>	2.30±1.33
	EDITS	84.42±2.87	3.74±3.54	4.46±3.50	OOM	OOM	OOM	OOM	OOM	OOM
	NIFTY	84.11±5.49	5.74±0.38	4.07±1.28	68.48±1.11	3.84±1.05	3.90±2.18	66.68±1.45	6.75±1.84	8.15±0.97
	FairVGNN	88.41±1.29	1.14±0.67	1.69±1.13	68.50±0.71	<u>1.12±0.98</u>	1.13±1.02	66.39±1.95	4.15±1.30	2.31±1.57
	FairSIN-G	88.79±1.08	3.97±0.92	<u>1.70±0.66</u>	69.11±0.62	2.00±1.13	1.66±0.70	66.19±1.49	4.96±0.25	2.90±1.21
	FairSIN-F	88.51±0.16	0.67±0.33	1.85±0.50	<u>69.28±0.98</u>	1.80±0.46	1.62±0.84	66.99±1.06	3.25±1.00	1.89±0.79
	FairSIN w/o N.	87.70±0.28	<u>0.64±0.40</u>	2.21±0.22	68.77±0.62	2.35±0.99	1.71±0.99	67.39±1.05	2.92±1.69	1.79±1.16
	FairSIN w/o D.	88.46±0.19	0.82±0.51	2.12±0.55	69.65±0.32	1.91±0.82	<u>1.09±1.12</u>	66.78±0.83	3.92±1.02	<u>1.62±0.68</u>
	FairSIN	88.74±0.42	0.58±0.60	1.49±0.34	69.12±1.16	1.04±0.83	1.04±0.42	67.95±0.79	1.74±0.73	0.68±0.65

Table 2: Comparison among SOTA methods and different variants of FairSIN. (Bold: the best; underline: the runner-up.)

FairSIN: Experiment

❖ Limitation

- ❑ Message passing enlarges the sensitive biases for both raw and neutralized features
 - Validating the theoretical analysis
- ❑ Neutralized features have much less sensitive information leakage
 - Demonstrating the effectiveness of F3
- ❑ MLP_{ϕ}^k can provide additional information when calculating representations
 - Especially for the nodes with few heterogeneous neighbors

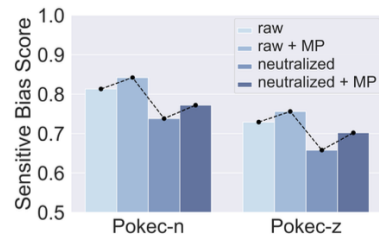


Figure 3: Sensitive biases in four groups of features. The biases are measured by average $\hat{P}_{\theta}(s|x)$, and larger scores indicate more serious sensitive leakage in the representations.

Paper \ Problems	Attribute Bias	Structural Bias	Potential Bias
	0	X	Δ
FairSIN	0	X	Δ

One-size-fits-all

1. FairGNN
2. FairVGNN
3. FairSIN

The Specialists

4. EDITS
5. DAB-GNN



EDITS: Modeling and Mitigating Data Bias for Graph Neural Networks

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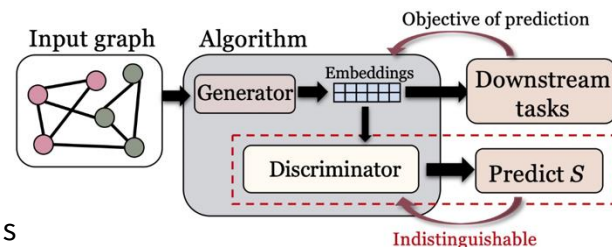
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WWW'22

EDITS: Model-agnostic Attributed Network Debiasing

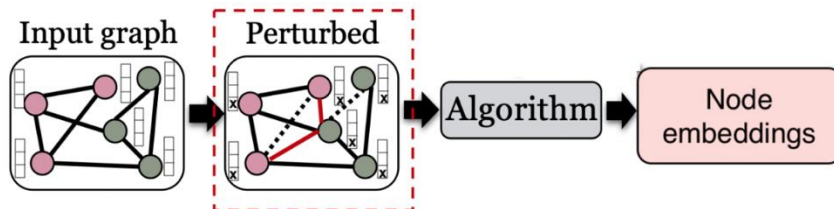
❖ Motivation

- ❑ Existing approaches to debias GNN outcomes, for a specific GNN model on a certain downstream task
 - **Generator:** generate node embeddings for downstream tasks
 - **Discriminator:** distinguish the embeddings between demographic subgroups
- ❑ Considering that debiasing for different GNNs is expensive



❖ Present Work

- ❑ Desire to mitigate the bias in attribute networks before being fed to GNNs
- ❑ Model-agnostic manner: independent of any specific GNN



EDITS: Model-agnostic Attributed Network Debiasing

❖ Propose definitions and metrics to measure the bias in an attribute network

❑ Attribute Bias

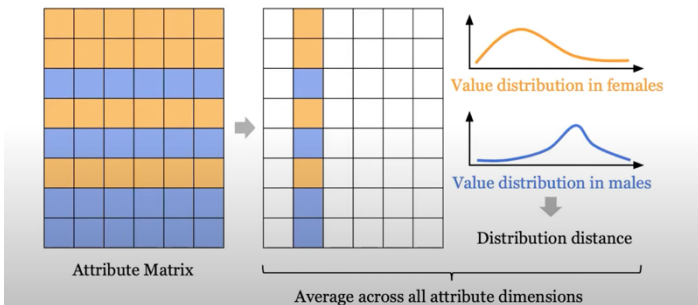
- For any attribute, if its value distributions between different demographic groups are different, then attribute bias exists in G

❑ Structural Bias

- For the attribute values propagated w.r.t. A, if their distributions between different demographic groups are different at any attribute dimension, then structural bias exists in G

Bias Metrics: attribute bias

$$X_{\text{norm}} \in \mathbb{R}^{N \times M} \rightarrow X_{\text{total}} = \{(\mathcal{X}_1^0, \mathcal{X}_1^1), (\mathcal{X}_2^0, \mathcal{X}_2^1), \dots, (\mathcal{X}_M^0, \mathcal{X}_M^1)\} \rightarrow b_{\text{attr}} = \frac{1}{M} \sum_m W(\text{pdf}(\mathcal{X}_m^0), \text{pdf}(\mathcal{X}_m^1))$$



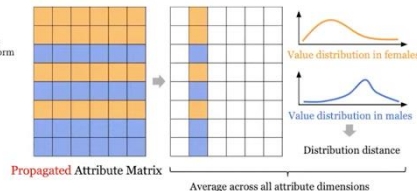
Bias Metrics: structural bias

$$P_{\text{norm}} = \alpha A_{\text{norm}} + (1 - \alpha) I \quad M_H = \beta_1 P_{\text{norm}} + \beta_2 P_{\text{norm}}^2 + \dots + \beta_H P_{\text{norm}}^H$$

$$R = M_H X_{\text{norm}}$$

$$\mathcal{R}_{\text{total}} = \{(\mathcal{R}_1^0, \mathcal{R}_1^1), (\mathcal{R}_2^0, \mathcal{R}_2^1), \dots, (\mathcal{R}_M^0, \mathcal{R}_M^1)\}$$

$$b_{\text{stru}} = \frac{1}{M} \sum_m W(\text{pdf}(\mathcal{R}_m^0), \text{pdf}(\mathcal{R}_m^1))$$



	Attribute Bias		Structural Bias	
	Vanilla	EDITS	Vanilla	EDITS
Pokec-z	0.43	0.33 (−23.3%)	0.83	0.75 (−9.64%)
Pokec-n	0.54	0.42 (−22.2%)	1.03	0.89 (−13.6%)
UCSD34	0.53	0.48 (−9.43%)	0.68	0.63 (−7.35%)
German	6.33	2.38 (−62.4%)	10.4	3.54 (−66.0%)
Credit	2.46	0.56 (−77.2%)	4.45	2.36 (−47.0%)
Recidivism	0.95	0.39 (−58.9%)	1.10	0.52 (−52.7%)

EDITS

❖ Proposed Debiasing Framework (*Parallel Debiasing & Objective*)

- ❑ **Attribute Debiasing** : Learn a debiasing function g_θ with learnable parameter $\theta \in \mathbb{R}^M \rightarrow \hat{X} = g_\theta(X)$
- ❑ **Structural Debiasing** : $\hat{A}$, initialized with A at the beginning
- ❑ **Wasserstein Distance Approximator**: Learn a function f for each attribute dimension
 - Utilized to estimate the W between the distributions of different groups for any attribute dimension

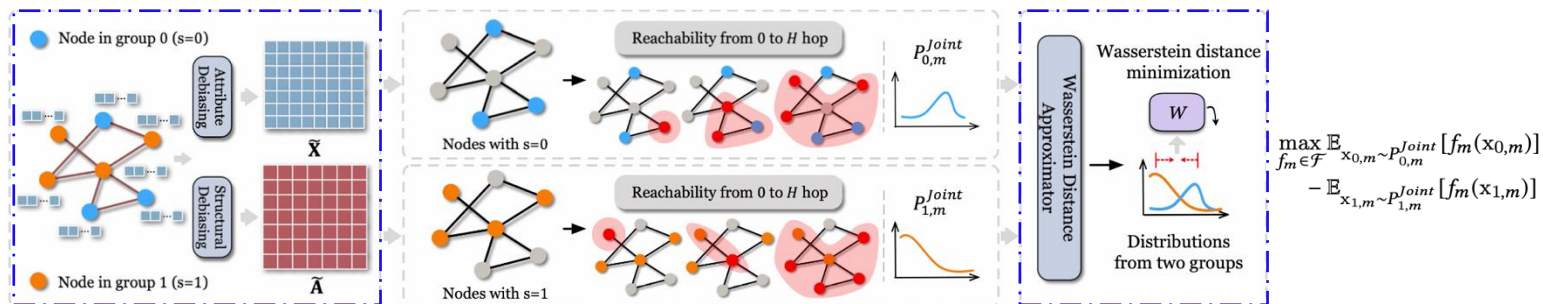


Figure 2: An illustration of EDITS with $H = 2$: Wasserstein Distance Approximator yields the approximated Wasserstein distance between $P_{0,m}^{Joint}$ and $P_{1,m}^{Joint}$; Attribute Debiasing and Structural Debiasing are optimized towards less biased $\tilde{X}$ and $\tilde{A}$.

EDITS: Experiments

❖ Observation

- EDITS achieves comparable model utility with alternatives but exhibits better bias mitigation performance

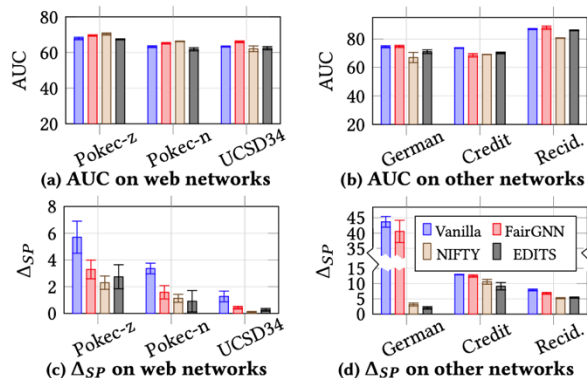


Figure 3: Performance comparison between EDITS and baselines on utility (AUC) and bias mitigation (Δ_{SP}).

Paper \ Problems	Attribute Bias	Structural Bias	Potential Bias
	0	0	x



Disentangling, Amplifying, and Debiasing: Learning Disentangled Representations for Fair Graph Neural Networks

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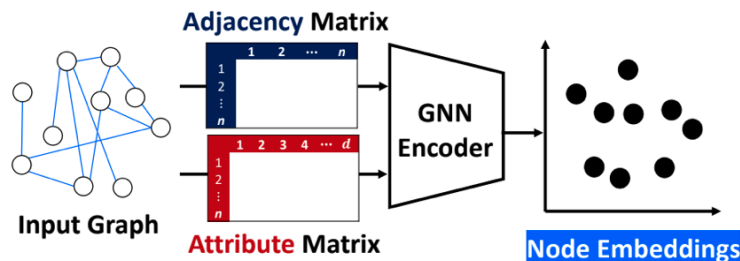
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AAAI'25

DAB-GNN

❖ Motivation

- ❑ Predictions based on node embeddings learned by GNNs can be unfair due to the biases inherent in the input graph data

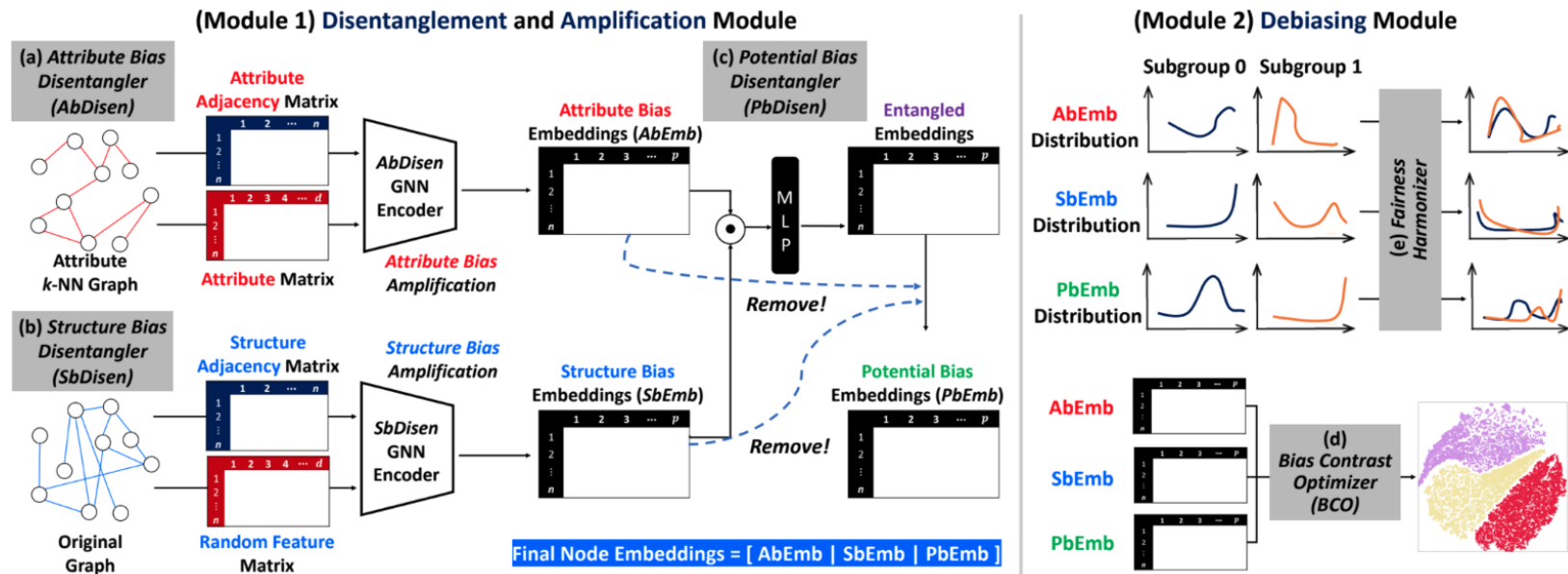


- **Attribute Bias:** Node attributes may exhibit different distributions across subgroups defined by sensitive attributes (e.g., gender)
- **Structure Bias:** Nodes within similar sensitive attributes tend to form connections
- **Potential Bias:** When node attributes and graph structure interact in the GNN mechanism, new biases may emerge that were not evident in either aspect alone

- ❑ However, existing fairness-aware GNN methods try to **tackle all biases at once** by using **a single, entangled embedding for each node**

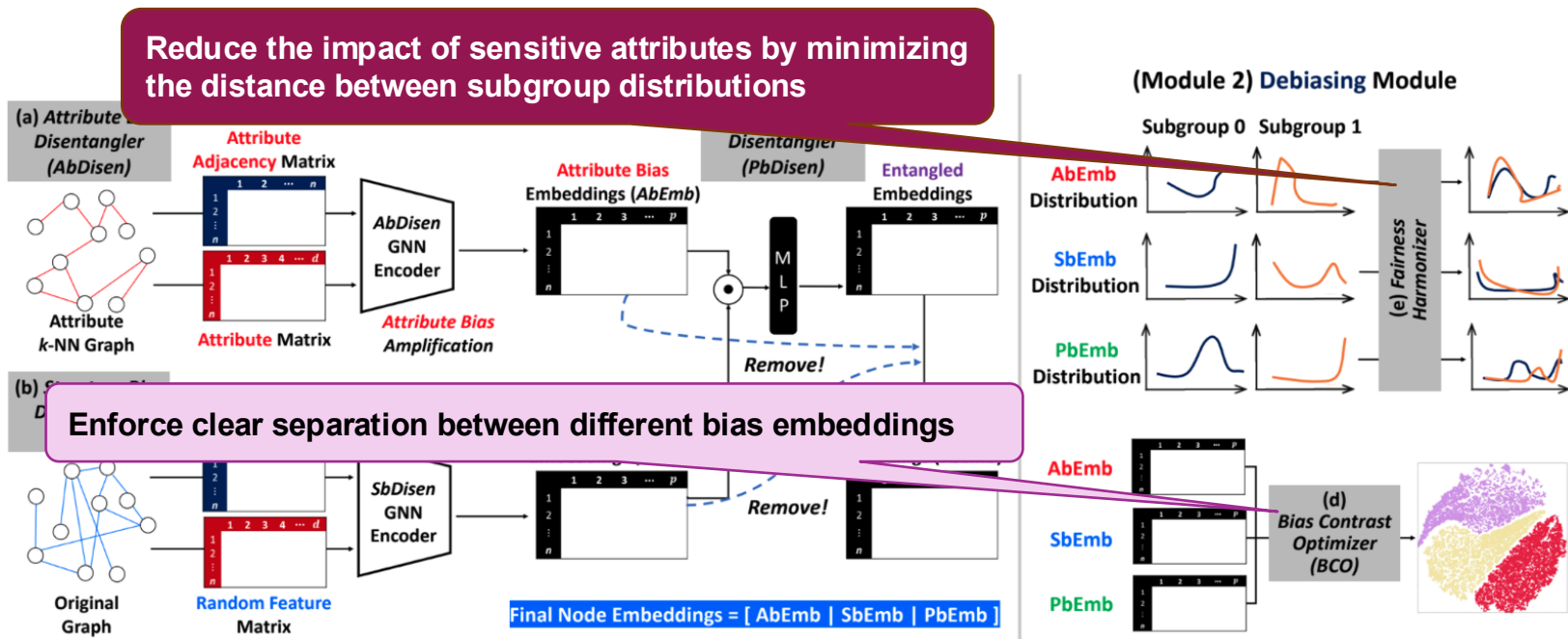
DAB-GNN: Framework

- ❖ Disentangles, Amplifies, and deBiases the attribute, structure, and potential biases through a GNN framework



DAB-GNN: Framework

- ❖ Disentangles, Amplifies, and deBiases the attribute, structure, and potential biases through a GNN framework



DAB-GNN: Experiments

❖ Observation

- ❑ Not only address fairness effectively but also enhance accuracy
 - A few cases have slightly lower accuracy/higher fairness values
 - CAF : Underscore difficulty in balancing accuracy and fairness

	Metrics	L1-Vanilla	L3-Vanilla	FairGNN	NIFTY	EDITS	FairVGNN	CAF	GEAR	BIND	PFR-AX	PostProcess	FairSIN	DAB-GNN
NBA	ACC (↑)	57.97	58.73	60.76	63.29	69.11	65.57	60.51	57.98	60.76	<u>70.63</u>	58.73	66.58	71.39
	AUC (↑)	63.75	63.33	74.91	70.75	71.82	79.96	67.06	60.04	79.33	73.26	63.33	71.72	80.56
	F1 (↑)	61.55	62.00	70.69	66.86	74.99	72.93	68.81	65.08	70.50	74.06	62.00	<u>74.21</u>	73.51
	SP (↓)	32.94	32.83	6.39	9.82	8.98	7.82	0.00	20.53	4.55	4.03	32.83	12.96	1.12
	EO (↓)	33.68	35.95	10.14	8.60	4.39	13.28	0.00	21.94	1.77	13.56	35.95	2.34	<u>0.80</u>
Recidivism	ACC (↑)	84.18	83.73	84.50	79.94	78.18	83.64	<u>86.79</u>	78.32	84.49	85.41	81.28	86.59	89.99
	AUC (↑)	86.90	86.84	89.05	81.23	83.62	84.38	87.07	81.30	89.13	<u>89.48</u>	83.23	89.08	93.41
	F1 (↑)	78.65	78.10	79.77	69.77	73.16	76.89	80.63	71.18	79.82	79.48	75.91	<u>80.87</u>	86.31
	SP (↓)	7.79	8.13	6.64	3.69	10.89	5.42	5.73	5.81	9.24	6.13	<u>1.43</u>	5.65	0.73
	EO (↓)	5.23	5.65	3.16	2.97	7.62	3.92	3.41	4.11	4.61	4.14	<u>2.92</u>	3.59	0.90
Credit	ACC (↑)	73.57	73.92	73.99	73.43	74.77	<u>77.92</u>	76.00	<u>o.o.m</u>	74.60	63.96	73.21	77.60	78.19
	AUC (↑)	73.48	<u>73.40</u>	64.19	72.14	72.30	68.67	65.72	<u>o.o.m</u>	71.91	66.90	70.10	71.57	71.41
	F1 (↑)	81.87	82.16	83.08	81.70	82.99	87.48	85.15	<u>o.o.m</u>	82.76	73.95	82.03	87.23	<u>87.39</u>
	SP (↓)	13.88	12.18	3.17	11.60	7.98	0.40	11.70	<u>o.o.m</u>	11.76	19.19	1.39	0.69	<u>0.44</u>
	EO (↓)	11.68	10.04	1.73	9.30	6.09	0.16	8.51	<u>o.o.m</u>	9.15	22.66	1.83	0.66	<u>0.45</u>
Pokec_n	ACC (↑)	66.97	65.27	63.56	<u>67.86</u>	<u>o.o.m</u>	69.51	<u>o.o.m</u>	<u>o.o.m</u>	55.69	<u>o.o.m</u>	66.54	65.69	67.18
	AUC (↑)	72.73	70.74	67.10	<u>73.92</u>	<u>o.o.m</u>	73.99	<u>o.o.m</u>	<u>o.o.m</u>	58.99	<u>o.o.m</u>	71.76	72.89	73.68
	F1 (↑)	65.70	64.91	59.79	<u>66.25</u>	<u>o.o.m</u>	66.01	<u>o.o.m</u>	<u>o.o.m</u>	52.36	<u>o.o.m</u>	65.91	67.44	62.34
	SP (↓)	7.90	17.19	3.28	<u>1.20</u>	<u>o.o.m</u>	2.77	<u>o.o.m</u>	<u>o.o.m</u>	6.78	<u>o.o.m</u>	14.97	2.40	0.71
	EO (↓)	7.09	14.88	5.05	<u>1.23</u>	<u>o.o.m</u>	3.38	<u>o.o.m</u>	<u>o.o.m</u>	5.96	<u>o.o.m</u>	11.38	1.64	1.09
Pokec_z	ACC (↑)	64.92	65.40	62.97	<u>65.71</u>	<u>o.o.m</u>	63.38	<u>o.o.m</u>	<u>o.o.m</u>	58.38	<u>o.o.m</u>	64.39	62.21	68.56
	AUC (↑)	70.03	69.84	65.81	<u>70.57</u>	<u>o.o.m</u>	68.99	<u>o.o.m</u>	<u>o.o.m</u>	61.20	<u>o.o.m</u>	69.08	68.81	74.85
	F1 (↑)	65.48	65.08	64.47	65.00	<u>o.o.m</u>	<u>67.31</u>	<u>o.o.m</u>	<u>o.o.m</u>	58.13	<u>o.o.m</u>	65.45	65.37	67.94
	SP (↓)	7.27	10.91	4.79	5.03	<u>o.o.m</u>	5.04	<u>o.o.m</u>	<u>o.o.m</u>	6.13	<u>o.o.m</u>	12.18	<u>0.96</u>	0.67
	EO (↓)	4.05	7.88	3.65	<u>1.24</u>	<u>o.o.m</u>	3.06	<u>o.o.m</u>	<u>o.o.m</u>	4.96	<u>o.o.m</u>	7.14	1.64	0.73

Table 4: Accuracy and fairness results of **DAB-GNN** and competitors across five real-world datasets. (↑) and (↓) mean higher and lower values are better, respectively; 'o.o.m' denotes 'out of memory.'

DAB-GNN: Experiments

❖ Disentangled Embeddings Analysis

- ❑ Different colors indicate different types of bias embeddings
- ❑ Bias embeddings are well-separated into distinct clusters
- ❑ Successfully isolates various biases present in graph data

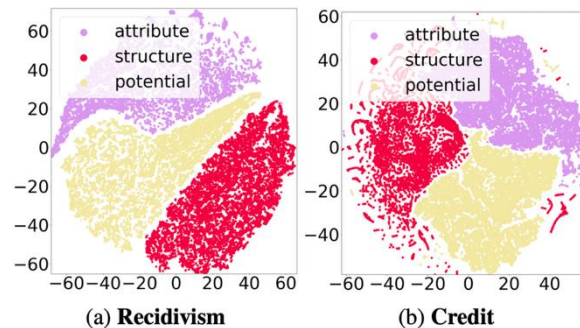


Figure 2: Visualization of disentangled node embeddings by using t-SNE: *AbEmb*, *SbEmb*, and *PbEmb*.

Paper \ Problems	Attribute Bias	Structural Bias	Potential Bias
	0	0	0

Conclusion

Graph Representation Learning

❖ **Graph:** A general language for describing and analyzing **entities** with **relations/interactions**

❖ **Representation Learning:** Automatically learn node feature

❖ **Graph Neural Networks**

- Generate node embeddings based on local network neighborhoods

- **(1) Message:** each node computes a message

$$\mathbf{m}_u^{(l)} = \text{MSG}^{(l)}(\mathbf{h}_u^{(l-1)}), u \in \{N(v) \cup v\}$$

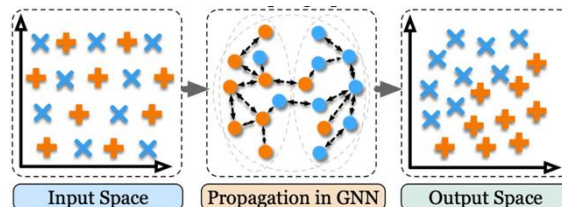
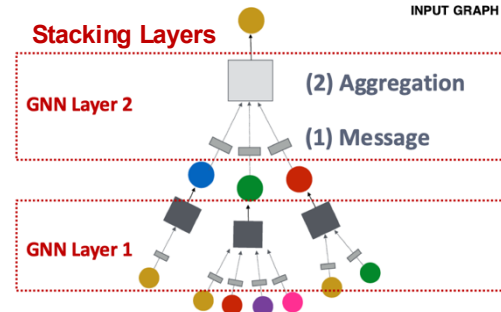
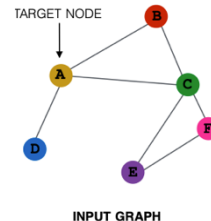
- **(2) Aggregation:** aggregate messages from neighbors

$$\mathbf{h}_v^{(l)} = \text{AGG}^{(l)}(\{\mathbf{m}_u^{(l)}, u \in N(v)\}, \mathbf{m}_v^{(l)})$$

- Enrich node representation and preserve both node feature characteristics and topological structures

❖ **GNNs trained on graphs may inherit the societal bias in data**

- Limit adoption of GNNs in many real-world applications
- Important to investigate fair GNNs
- The bias can be revealed as there:
 - Attribute bias, structural bias, and potential bias



An example in Graph Neural Networks (GNNs): the unbalance between intra-group and inter-group edges could easily induce bias in the outcome space [1].

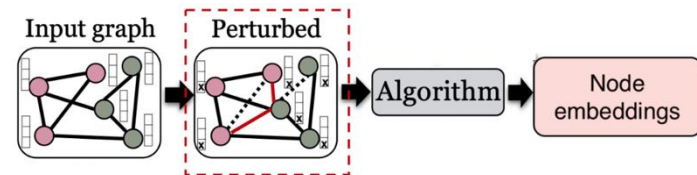
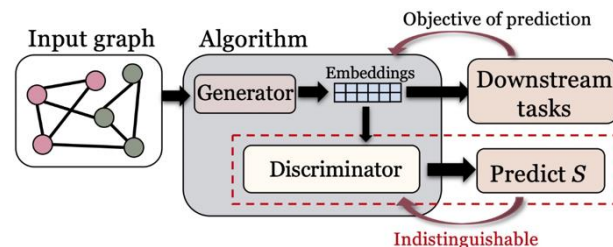
Fairness-aware Graph Learning

❖ Fairness Definition: Lack of favoritism from one side or another

- ❑ SP – Equal acceptance rate
- ❑ EO – Equal true positive rate

❖ Findings

- ❑ One-size-fits-all methods (Adversarial Debiasing)
 - FairGNN : Fairness with **Limited** Sensitive Attribute Information
 - FairVGNN : Fair View to Prevent Sensitive Information **Leakage**
 - FairSIN : Fair GNNs through Sensitive Information **Neutralization**
- ❑ The Specialists methods (Data Perturbation)
 - EDITS : Model-agnostic Attributed Network Debiasing
 - DAB-GNN : Disentangles, Amplifies, and debiases the three-bias



Appendix

❖ Acknowledgements

- ❑ Part of the slides are credited to the following
 - Jian Kang (MBZUAI - Mohamed bin Zayed University of Artificial Intelligence)
 - Yushun Dong (Florida State University)
 - CS224W: Machine Learning with Graphs (<https://web.stanford.edu/class/cs224w>)

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Appendix

❖ References

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Thank You!



"Output is conditional on INPUT."